

Class 9 Social Science — Chapter 3

Atmosphere and Climate

1. ATMOSPHERE

Meaning of Atmosphere

The **atmosphere** is the blanket of air surrounding the Earth.

Earth's **gravity** keeps this blanket of air around the planet.

The atmosphere is not made of only one gas. It is a **mixture of many gases**, along with water vapour and tiny dust particles.

Why is the atmosphere important?

The atmosphere is essential because it:

- provides gases needed for life;
- protects Earth from harmful radiation from the Sun;
- filters dangerous ultraviolet radiation;
- helps regulate Earth's temperature;
- contains water vapour needed for clouds and rainfall;
- creates weather conditions;
- influences temperature, humidity and air pressure.

Simple idea

Think of the atmosphere as Earth's **protective blanket**.

Just as a blanket protects the body and reduces heat loss, the atmosphere protects the Earth and helps control its temperature.

2. GRAVITY AND THE ATMOSPHERE

Gravity is the force of attraction between objects having mass.

Earth's gravity pulls objects towards Earth. It also helps keep the atmosphere around our planet instead of allowing all the gases to escape into space.

Difficult word

Gravity = force that pulls objects towards the Earth.

3. COMPOSITION OF THE ATMOSPHERE

The atmosphere contains different gases in different proportions.

The two major gases are:

Gas	Approximate Amount
Nitrogen	78%
Oxygen	21%
Argon	0.93%
Carbon dioxide	0.04%
Other gases	0.03%

Other gases include:

- helium,
- neon,
- krypton,
- xenon,
- ozone,
- hydrogen.

The atmosphere also contains:

- **water vapour**, and
- **tiny dust particles**.

3.1 Water Vapour in the Atmosphere

Water vapour generally forms about **0.1% to 0.4%** of the atmosphere.

Its amount is not the same everywhere.

Water vapour is very important because it helps in:

- cloud formation;
- precipitation;
- rainfall;
- humidity.

The composition of the atmosphere also changes with **altitude**.

Difficult word

Altitude = height of a place above mean sea level.

Mean sea level is taken as **zero**.

4. STRUCTURE OF THE ATMOSPHERE

The atmosphere is divided into different layers.

These layers are mainly identified by changes in:

- **temperature**, and
- **air density**

as altitude increases.

Air is densest near the Earth's surface. Its density decreases as we go higher.

The five major layers are:

Earth → Troposphere → Stratosphere → Mesosphere → Thermosphere → Exosphere → Space

5. TROPOSPHERE

The **troposphere** is the lowest and most important layer of the atmosphere.

Main features

- Its average height is about **12 km**.
- Temperature **decreases** as altitude increases.
- The air that we breathe is found here.
- Most atmospheric water vapour is found here.
- Most clouds form here.
- Almost all weather phenomena occur here.

Examples of weather phenomena:

- rainfall,
- fog,
- hail,
- cloud formation.

The boundary between the troposphere and stratosphere is called the **tropopause**.

“In which layer does weather occur?”

The answer is **troposphere**.

6. STRATOSPHERE

The **stratosphere** lies above the troposphere.

It extends up to about **50 km**.

Main features

- It is comparatively free from clouds.
- It has fewer weather disturbances.
- Therefore, it is suitable for aeroplanes.
- It contains the important **ozone layer**.
- The ozone layer filters harmful solar radiation, including ultraviolet radiation.

The boundary between the stratosphere and mesosphere is called the **stratopause**.

Why is ozone important?

Ozone protects living organisms by filtering harmful radiation coming from the Sun.

Especially important is **ultraviolet radiation**, often written as **UV radiation**.

Easy memory

Stratosphere = aeroplanes + ozone

7. MESOSPHERE

The **mesosphere** lies above the stratosphere.

It extends up to about **80 km**.

Main features

- Temperature decreases as altitude increases.
- Most meteorites entering Earth's atmosphere burn up in this layer.

Important fact

According to the chapter, temperature decreases with altitude particularly in:

- **Troposphere**
- **Mesosphere**

Easy memory

Mesosphere = meteors

8. THERMOSPHERE

The thermosphere extends roughly from **80 km to 700 km**.

Main features

In this layer:

- temperature rises very rapidly with altitude;
- gas molecules absorb X-rays and short-wave ultraviolet radiation from the Sun;
- the **ionosphere** forms a part of the thermosphere;
- radio transmission is helped by the reflection of radio waves;
- auroras occur here.

Difficult word

Ionosphere = a region within the thermosphere that helps in radio communication.

9. AURORAS

An **aurora** is a colourful display of light in the sky near the polar regions.

The word *aurora* means something like **dawn or morning light**.

How does an aurora form?

The Sun releases charged particles called **solar wind**.

These particles reach Earth's atmosphere.

Earth's magnetic field directs many of them towards the magnetic poles.

When the particles interact with atmospheric gases, the gases glow in different colours.

Names

- **Aurora Borealis** → Northern Hemisphere
- **Aurora Australis** → Southern Hemisphere

10. EXOSPHERE

The **exosphere** is the uppermost layer of the atmosphere.

Main features

- Air is extremely thin.
- Gravity is weak.
- Light gases such as **hydrogen and helium** may escape into space.

Easy memory

Exosphere = exit towards space

11. QUICK COMPARISON OF ATMOSPHERIC LAYERS

Layer	Main Features
Troposphere	Weather, clouds, water vapour, breathable air
Stratosphere	Ozone layer, suitable for aeroplanes
Mesosphere	Meteorites burn here
Thermosphere	Very high temperature, ionosphere, auroras
Exosphere	Uppermost layer, extremely thin air

12. WHY DO WE NOT FEEL ATMOSPHERIC PRESSURE ON OUR BODY?

Air has weight.

Therefore, the atmosphere puts pressure on our body.

Yet we normally do not feel this pressure because:

- atmospheric air presses on us from all directions;
- our body also produces an internal counter-pressure;
- these pressures balance one another.

13. WEATHER AND CLIMATE

Students often confuse these two terms.

Weather

Weather means the hour-to-hour or day-to-day condition of the atmosphere.

For example:

- today is rainy;

- this afternoon is windy;
- tomorrow may be sunny.

Weather can change quickly.

Climate

Climate means the average weather conditions and variations of a place over a long period. The chapter generally considers a period of **30 years or more** for understanding climate.

Weather vs Climate

Weather	Climate
Short-term condition	Long-term pattern
Changes hourly or daily	Observed over many years
Covers a particular time	Represents general conditions
Example: It is raining today.	Example: A region has a hot and humid climate.

Golden rule

Weather = What is happening now?

Climate = What usually happens there?

14. ELEMENTS OF WEATHER AND CLIMATE

Five major elements are discussed in this chapter:

1. **Temperature**
2. **Humidity**
3. **Precipitation**
4. **Atmospheric pressure**
5. **Wind**

All five are connected.

For example:

Temperature changes → pressure changes → wind develops → moisture moves → rainfall may occur

This relationship is extremely important for understanding the monsoon.

15. TEMPERATURE

Atmospheric temperature changes:

- between day and night;
- from season to season;
- from one region to another.

Summer is generally warmer than winter.

One important factor controlling temperature is **insolation**.

Insolation

Insolation means the incoming solar energy from the Sun that is intercepted by Earth.

Insolation and latitude

The amount of insolation generally decreases from the **equator towards the poles**.

Therefore:

More direct sunlight near Equator → more heating

Less direct sunlight towards poles → less heating

So temperature generally decreases from the equator towards the poles.

16. TEMPERATURE ZONES OF THE EARTH

Torrid Zone

This region lies around the equator.

Sunlight is more direct here.

Therefore, it generally receives stronger heating.

Temperate Zones

These lie beyond the tropical region towards the north and south.

They receive less direct sunlight.

Frigid Zones

These lie towards the poles.

Sunlight reaches them at a low angle.

Therefore, they are much colder.

The figure identifies important latitudes such as:

- Equator — 0°
- Tropic of Cancer — $23\frac{1}{2}^{\circ}$ N
- Tropic of Capricorn — $23\frac{1}{2}^{\circ}$ S
- Arctic Circle — $66\frac{1}{2}^{\circ}$ N
- Antarctic Circle — $66\frac{1}{2}^{\circ}$ S.

17. HUMIDITY

Water from:

- rivers,
- lakes,
- seas,
- oceans,
- land surfaces

evaporates and enters the atmosphere as **water vapour**.

The presence of water vapour in the air creates **humidity**.

Therefore:

Humidity = moisture or water vapour present in the air.

Warm air can hold more water vapour.

When humidity is high:

- clothes take longer to dry;
- sweat evaporates slowly;
- we feel uncomfortable.

Everyday example

After heavy rain on a hot day, the air may feel sticky.

This happens because the air contains a large amount of moisture.

18. PRECIPITATION

When air becomes saturated with water vapour, the vapour may condense.

The water then falls towards Earth because of gravity.

This falling of moisture is called **precipitation**.

Forms of precipitation

- drizzle,
- rain,
- snow,
- sleet,
- hail.

Factors affecting precipitation

- prevailing winds,
- mountains,
- seasons.

When precipitation falls in liquid form, it is called **rain**.

Importance of Rain

Rain:

- is the most common form of precipitation;
- can lower the temperature of a place;
- helps replenish groundwater;
- supports agriculture;
- provides water for people and ecosystems.

A very long absence of rainfall leads to a **dry climate**.

19. ATMOSPHERIC PRESSURE

Air has weight.

The pressure produced by the weight of air on Earth's surface is called **atmospheric pressure** or **air pressure**.

Pressure and altitude

Air pressure is:

highest at sea level

and

decreases as altitude increases.

20. TEMPERATURE AND AIR PRESSURE

This connection is extremely important.

Hot area

When air becomes hot:

1. air heats up;
2. it becomes lighter;
3. it rises;
4. low pressure develops near the surface.

Therefore:

Hot air rises → Low pressure

Cold area

When air becomes cold:

1. it becomes heavier;
2. it sinks;
3. high pressure develops.

Therefore:

Cold air sinks → High pressure

Pressure and Weather

Low Pressure

Low pressure is commonly connected with:

- clouds;
- wet weather.

High Pressure

High pressure is commonly connected with:

- clear skies;
 - sunny weather.
-

21. WIND

Wind is the movement of air from a high-pressure area to a low-pressure area.

This is one of the most important rules in the chapter.

Remember:

HIGH PRESSURE → LOW PRESSURE

Air does **not normally move from low pressure to high pressure**.

The direction of wind is also influenced by Earth's rotation.

22. HOW ARE WINDS NAMED?

A wind is named according to the direction **from which it comes**.

Example:

If a wind comes **from the west**, it is called a **westerly** wind.

23. WIND SPEED AND ITS EFFECTS

The textbook gives examples of how different wind speeds affect our surroundings.

Type	Speed	Common Effect
Calm	0–1 km/h	Smoke rises vertically
Light breeze	6–11 km/h	Felt on face; leaves rustle
Strong breeze	39–49 km/h	Large branches sway; umbrellas become difficult to use
Storm	103–117 km/h	May cause widespread damage

24. LOCAL WINDS

Two important local winds discussed in the chapter are:

- **Sea breeze**
- **Land breeze**

They help create more moderate climatic conditions in coastal areas.

25. SEA BREEZE

A **sea breeze** is a local wind that blows:

Sea → Land

It mainly occurs during the **day**, especially in the afternoon.

How does it form?

During the day:

1. the Sun heats both land and water;
2. land heats faster than water;
3. land becomes hotter;
4. air above the land becomes warm and rises;
5. low pressure develops over the land;
6. the sea remains relatively cooler;
7. air moves from the sea towards the land.

Therefore:

Day → Hot land → Low pressure → Sea to land → Sea breeze

26. LAND BREEZE

A **land breeze** is a local wind that blows:

Land → Sea

It occurs mainly during the **night**.

How does it form?

At night:

1. land loses heat faster than water;
2. land becomes cooler than the sea;
3. pressure differences develop;
4. air moves from land towards the sea.

The chapter also notes that the temperature and pressure difference is smaller, so land-breeze wind speed is usually low.

27. SEA BREEZE VS LAND BREEZE

Sea Breeze	Land Breeze
Occurs mainly during day	Occurs mainly during night
Land is warmer	Land is cooler
Blows sea → land	Blows land → sea
Low pressure develops over warm land	Land cools faster than sea
Especially common in afternoon	Usually weaker

Best memory trick

Sea breeze → Sea sends air to land

Land breeze → Land sends air to sea

28. SEASONS

Weather may change from one day to another.

However, certain weather conditions continue in a general pattern for several weeks or months.

Examples:

- hot or cool,
- wet or dry,
- windy or calm,
- cloudy or bright.

On the basis of these repeating patterns, a year is divided into **seasons**.

29. CLIMATE OF INDIA

India's climate can broadly be described as a:

Tropical Monsoon Climate

The **Indian Meteorological Department (IMD)** recognises four main seasons:

1. Winter
2. Summer or pre-monsoon
3. Monsoon or rainy season
4. Post-monsoon or retreating monsoon

30. WINTER SEASON

Winter generally lasts from:

December to early April

Important points

- December and January are the coldest months.
- In north-western India, average temperatures may be about **10–15°C**.
- Temperature increases towards the equator.
- In south-eastern mainland India, it may reach around **20–25°C**.

31. SUMMER OR PRE-MONSOON SEASON

It generally lasts from:

April to June

In north-western India, it may continue up to July.

Important points

- April is generally hottest in western and southern regions.
- May is generally hottest in northern regions.
- Much of inland India experiences average temperatures around **32–40°C**.

32. MONSOON OR RAINY SEASON

The advancing monsoon generally lasts from:

June to September

It is dominated by the humid **south-west monsoon**.

The south-west monsoon gradually spreads across the country from late May or early June.

Withdrawal begins from north India around the beginning of October.

33. POST-MONSOON OR RETREATING MONSOON

This season generally lasts from:

October to December

In north-western India, October and November are usually cloudless.

Parts of southern India receive important rainfall during this period.

34. ADDITIONAL SEASONS IN THE HIMALAYAS

The Himalayan states have a more **temperate (moderate)** climate.

They also experience:

- **autumn**
- **spring**

in addition to the main seasonal divisions.

35. TRADITIONAL SIX SEASONS OF INDIA

Traditionally, India recognises six seasons, each lasting roughly two months.

R̥tu	Indian Calendar Months	Gregorian Months
Vasanta	Chaitra–Vaiśākha	March–April
Gr̥iṣhma	Jyeshṭha–Āṣhāḍha	May–June
Var̥ṣhā	Śhrāvaṇa–Bhādrapada	July–August
Śharad	Āśhvina–Kārtika	September–October
Hemanta	Mārgaśhīrṣha–Pauṣha	November–December
Śhiśhira	Māgha–Phālguna	January–February

Their exact timing may differ from one region of India to another.

36. HISTORICAL UNDERSTANDING OF WEATHER IN INDIA

The chapter gives some interesting examples showing that people in India studied weather and rainfall long ago.

Kautilya's Arthaśhāstra

The **Arthaśhāstra** contains records of scientific measurements of rainfall.

Such information was also used for:

- revenue management;
- relief measures.

37. MONSOON

India's climate is strongly influenced by **monsoon winds**.

Meaning

A **monsoon** is the seasonal reversal of wind direction during a year.

The term comes from the Arabic word:

mausim = season

Ancient sailors noticed that winds reversed direction according to the season. They used these winds for sailing to and from India.

38. ANCIENT METHODS OF RAINFALL PREDICTION

Because agriculture depended strongly on rainfall, people developed methods of predicting the monsoon.

The chapter mentions:

- **Kṛīṣhiparāśhara**
- **Bṛīhatsamhitā**
- **Varāhamihira**

Kṛīṣhiparāśhara used the positions of the Moon and Sun.

Varāhamihira also considered **nakṣhatras (lunar mansions)** for predicting seasonal rainfall.

39. NAKṢHATRAS OR LUNAR MANSIONS

The chapter describes nakṣhatras as **27 divisions of the sky** along the Moon's path.

The Moon takes about 27 days to move around Earth relative to the stars.

Therefore, it appears to move through approximately one nakṣhatra each day.

40. TWO IMPORTANT MONSOON SYSTEMS

Monsoon winds can mainly be divided into:

1. South-West Monsoon

Also called the **summer monsoon**.

2. North-East Monsoon

Also called the **winter monsoon**.

Understanding their mechanism is one of the most important parts of the chapter.

41. MECHANISM OF SOUTH-WEST MONSOON

The south-west monsoon mainly operates between:

June and September

It brings most of India's annual rainfall.

Step-by-step mechanism

Step 1 — Summer heating

During summer, the Indian landmass becomes very hot.

Step 2 — Unequal heating

Land heats more quickly than the surrounding oceans.

Therefore:

- India = hotter
- Indian Ocean = relatively cooler

Step 3 — Pressure difference

Hot air over India rises.

Therefore, a **low-pressure area** develops over India.

The cooler Indian Ocean has relatively **high pressure**.

Step 4 — Movement of wind

Air always tends to move:

High Pressure → Low Pressure

Therefore, winds start blowing:

Ocean → Indian landmass

Step 5 — Moisture

These winds pass over:

- Indian Ocean,
- Arabian Sea,
- Bay of Bengal.

They carry large amounts of moisture.

Step 6 — Rainfall

When these moisture-rich winds reach India, they bring widespread rainfall.

Complete flow

Summer → Land heats faster → Low pressure over India → High pressure over ocean → Moist winds move sea to land → Rainfall

This is the basic mechanism of the **south-west monsoon**.

42. ADVANCING SOUTH-WEST MONSOON

The textbook includes a map showing the normal advance of the south-west monsoon across India.

The broad pattern is:

late May → **southern/island regions**

↓

early June → **southern mainland India**

↓

June → **monsoon spreads northward**

↓

late June–early July → **large parts of northern and north-western India**

The map gives normal dates ranging from around **22 May to 8 July** for different regions.

Important concept

The monsoon does **not** arrive everywhere in India on the same day.

It advances gradually.

43. RETREATING MONSOON

The monsoon also does not disappear from the entire country at once.

It begins withdrawing from northern and north-western India and gradually retreats southwards.

The textbook map shows normal retreat dates from around **17 September to 15 October** for different regions.

Easy understanding

Advancing monsoon: rain-bearing winds spread over India.

Retreating monsoon: those winds gradually withdraw.

44. MECHANISM OF NORTH-EAST MONSOON

The north-east monsoon occurs mainly from:

October to February

It is also called the **winter monsoon**.

Step-by-step mechanism

Step 1 — Winter cooling

During winter, the Indian landmass cools faster than the surrounding oceans.

Step 2 — Pressure develops

Cool land develops **high pressure**.

The sea remains relatively warmer and has **lower pressure**.

Step 3 — Wind direction reverses

Air moves:

Land → **Sea**

Therefore, cold and dry winds blow away from India.

Step 4 — Most of India remains relatively dry

Since these winds begin over land, they generally do not contain much moisture.

Therefore, most parts of India do not receive much rain from them.

Step 5 — Bay of Bengal changes the situation

When some of these winds pass over the **Bay of Bengal**, they collect moisture.

Step 6 — Rainfall on south-eastern coast

They then bring rain especially to:

- Tamil Nadu,
- Andhra Pradesh,
- parts of Karnataka.

Therefore, winter monsoon rainfall is very important for south-eastern India.

45. SOUTH-WEST VS NORTH-EAST MONSOON

South-West Monsoon	North-East Monsoon
Summer monsoon	Winter monsoon
Mainly June–September	Mainly October–February
Land becomes hotter	Land becomes colder
Low pressure over land	High pressure over land
Winds mainly move sea → land	Winds mainly move land → sea
Winds contain large moisture	Winds start relatively dry
Gives most of India's rainfall	Important for south-eastern coast

South-West Monsoon	North-East Monsoon
Associated with advancing monsoon	Closely connected with retreating/winter conditions

46. WHY IS MONSOON IMPORTANT FOR INDIA?

Monsoon is not only a weather event.

It affects India's:

- agriculture,
- food supply,
- water,
- economy,
- transport,
- employment,
- festivals,
- daily life,
- rural livelihoods.

Agriculture

Many farmers depend on monsoon rainfall for:

- sowing seeds;
- growing crops.

Water supply

A good monsoon helps fill:

- rivers,
- reservoirs,
- wells.

Food production

Good rainfall usually supports better agricultural production.

Rural employment

Since many rural occupations depend on agriculture, rainfall also influences employment.

Too much or too little rain

Excess rainfall → Flood

Weak monsoon → Drought

Therefore, monsoon greatly influences India's economy and people's livelihoods.

47. KĀLIDĀSA AND THE MONSOON

The chapter states that Kālidāsa's **Meghadūtam**, written around the 5th century CE, mentions:

- the onset of monsoon over central India;
- the movement or path of monsoon clouds.

This shows that monsoon behaviour was observed and described in Indian literature long ago.

48. NATIONAL MONSOON MISSION

The **National Monsoon Mission (NMM)** comes under the Ministry of Earth Sciences.

Its main objective is to improve **monsoon prediction** over different time periods.

Modern weather and climate prediction models are used for this purpose.

49. MISSION MAUSAM

Mission Mausam aims to strengthen India's position in weather and climate science.

Its goals include making India:

- **Weather Ready**
- **Climate Smart**

It aims to improve:

- weather observation,
- climate observation,
- modelling,
- forecasting.

The information can help sectors such as:

- agriculture,
- disaster management,
- rural development.

50. CLIMATE CHANGE

Climate change is described as one of the most urgent challenges facing the Earth.

Meaning

Climate change means long-term changes in weather patterns such as:

- temperature,
- rainfall,
- wind.

The chapter mainly connects present climate change with human activities.

51. HUMAN CAUSES OF CLIMATE CHANGE

Important causes mentioned are:

Burning of fossil fuels

Burning fuels releases greenhouse gases into the atmosphere.

Deforestation

Deforestation means large-scale cutting or removal of forests.

Industrial pollution

Industries release gases such as carbon dioxide.

These activities increase greenhouse gases in the atmosphere.

52. GREENHOUSE GASES

The chapter mentions greenhouse gases such as:

- carbon dioxide,
- methane,
- nitrous oxide,
- water vapour.

These gases trap heat in the atmosphere.

When their amount increases, more heat can remain trapped.

This contributes to rising global temperatures.

53. EFFECTS OF CLIMATE CHANGE

The chapter identifies several serious effects.

Natural effects

- more frequent floods;
- droughts;
- melting glaciers;
- rising sea levels;
- loss of biodiversity.

Effects on people

Climate change can affect:

- human health;
- agriculture;
- livelihoods;
- different sections of society, including women and children.

Difficult word

Biodiversity = variety of living organisms such as plants, animals and other forms of life.

54. HOW CAN CLIMATE CHANGE BE REDUCED?

The chapter stresses **collective action**.

Important steps include:

- reducing carbon footprints;
 - using renewable energy;
 - protecting forests;
 - adopting sustainable lifestyles;
 - saving energy;
 - planting trees;
 - using resources carefully.
-

55. CARBON FOOTPRINT

A **carbon footprint** means the total amount of greenhouse gases released into the atmosphere because of human activities.

These activities may include:

- energy use,
- transportation,

- production of goods,
- production of services.

Simple example

A person who frequently uses a private car, wastes electricity and throws away plastic may create a larger environmental impact than a person who walks, saves electricity and recycles.

56. HOW OUR DAILY HABITS AFFECT OUR CARBON FOOTPRINT

The textbook examines four everyday areas.

A. Transport

Lower impact:

- walking;
- cycling.

Medium impact:

- public transport;
- carpooling.

Higher impact:

- private cars even for short distances.

Very high impact in the activity:

- frequent air travel.

B. Electricity

Lower impact:

- switching off lights and fans when not needed.

Higher impact:

- frequently leaving appliances switched on.

C. Water

Lower impact:

- using water carefully.

Higher impact:

- leaving taps running;
- taking unnecessarily long showers;
- wasting water.

D. Waste and Plastics

Lower impact:

- reuse;
- recycle;
- avoid single-use plastics.

Higher impact:

- regularly throwing away plastic;
- not recycling.

57. CARBON-FOOTPRINT ACTIVITY SCORING

The textbook activity gives:

Choice	Points
Low impact	1
Medium impact	2
High impact	3
Very high impact	4

The student then identifies changes that can reduce the score and makes a **Climate Action Pledge**.

58. PUNJAB FLOODS 2025 — CASE STUDY

This is an important **CBSE competency-based section** because it connects climate, rainfall, rivers, human activities and disaster management.

In 2025, Punjab experienced severe floods following:

- heavy monsoon rainfall;
- swelling of important rivers.

The case study mentions:

- Satluj,
- Beas,
- Ravi,
- and later Ghaggar.

Floodwaters damaged:

- villages,
- agricultural fields,
- houses,
- roads,
- bridges,
- other infrastructure.

The disaster caused:

- economic loss,
- social disruption,
- environmental damage.

59. NATURAL CAUSES OF THE PUNJAB FLOODS

The floods were not caused by one single factor.

1. Very heavy monsoon rainfall

Punjab received unusually heavy monsoon rainfall.

2. Western disturbances

Western disturbances brought additional moisture and rainfall.

3. Rain in neighbouring mountain regions

Heavy rain also occurred in:

- Himachal Pradesh;
- Jammu & Kashmir.

4. Rivers were already carrying high water

Important rivers were already flowing at high levels:

- Satluj,
- Beas,
- Ravi,
- Ghaggar.

5. Extra water caused overflowing

Additional rainfall and water coming from the hills caused the rivers to overflow.

This led to widespread flooding.

60. HUMAN-MADE CAUSES THAT WORSENE THE FLOODS

The chapter also shows that disasters may become worse because of human actions.

Weak and old embankments

Old river embankments or **dhūsi bāndh** could not hold the rising water.

Building too close to rivers

People had constructed houses and farms close to river channels.

This reduced the natural area where floodwater could spread safely.

Silt accumulation

Silt and mud had accumulated in rivers and dams.

This reduced their capacity to:

- hold water;
- carry water.

Poor warning

In some places:

- flood warnings came late;
- warnings were not communicated clearly.

As a result, people had less time to prepare.

All these factors increased the severity of the disaster.

61. EFFECTS OF THE PUNJAB FLOODS

The flood affected people, agriculture, animals, infrastructure and health.

Human impact

- People lost their lives.
- Thousands had to leave their homes.
- Many moved to relief camps.

Agricultural impact

- Large areas of farmland were flooded.
- Paddy crops suffered severe damage.

Livestock impact

- Poultry farms were damaged.
- Dairy farms were affected.
- Cows, buffaloes and chickens became sick or died.

Infrastructure impact

Damage occurred to:

- roads,
- bridges,
- border fences,
- public buildings.

Health impact

Dirty standing water resulted in:

- sanitation problems;
- spread of waterborne diseases.

62. MOST IMPORTANT LESSON FROM THE PUNJAB FLOOD CASE

A natural event does not always become a disaster because of nature alone.

The Punjab case shows:

Natural hazard + Human vulnerability = Greater disaster

Heavy rainfall was a natural cause.

But damage became worse because of:

- weak embankments;
- poor construction choices;
- reduced floodplain space;
- siltation;
- delayed warning.

This is an important **competency-based understanding** drawn directly from the case study.

63. CLIMATE DATA OF DIFFERENT INDIAN CITIES

The chapter provides representational monthly temperature and rainfall data for ten stations.

Among the stations visible in the table are:

- Bengaluru
- Mumbai
- Kolkata
- Delhi
- Jodhpur
- Chennai
- Nagpur
- Shillong
- Thiruvananthapuram
- Leh.

The purpose is not simply to memorise every figure.

Students should learn to **interpret climatic patterns**.

64. WHAT DOES THE CLIMATE TABLE TEACH US?

The data show that India has great climatic diversity.

For example:

Thiruvananthapuram

Temperatures stay around the mid-to-upper 20s °C through most of the year.

This shows a relatively small annual temperature range.

Delhi

Temperature changes strongly between winter and summer.

January is much cooler than May or June.

Jodhpur

Rainfall is comparatively low, with annual rainfall shown as about **36.6 cm**.

Shillong

Rainfall is very high, with annual rainfall shown as about **225.3 cm**.

Leh

Leh has:

- very cold winters;
- low annual precipitation;
- a high altitude of about **3506 m**.

Its annual rainfall/precipitation figure in the table is only about **8.5 cm**.

65. WHY DOES CHENNAI RECEIVE IMPORTANT RAINFALL LATER IN THE YEAR?

Look at Chennai's rainfall figures:

- October: about **30.6 cm**
- November: about **35.0 cm**

These values are much higher than many of its summer-month values.

This agrees with the chapter's explanation of the **north-east monsoon**.

North-east monsoon winds pass over the Bay of Bengal, collect moisture and bring rainfall to the south-eastern coast.

Concept connection

Retreating/winter monsoon + Bay of Bengal moisture → rainfall on south-eastern coast

66. WHY DOES THIRUVANANTHAPURAM HAVE AN EQUABLE CLIMATE?

Equable means having only small temperature differences.

From the table:

- January temperature: about **26.7°C**
- April temperature: about **28.7°C**
- July temperature: about **26.2°C**
- December temperature: about **26.5°C**

So temperature does not change greatly during the year.

The table clearly shows the equable pattern.

Additional CBSE context: Its coastal location helps reduce extreme heating and cooling because the sea heats and cools more slowly than land.

67. WHY DOES SHILLONG RECEIVE MUCH MORE RAINFALL THAN MANY OTHER STATIONS?

The table shows:

- Shillong annual rainfall: **225.3 cm**
- Kolkata annual rainfall: **162.5 cm**.

The chapter itself asks students to explain this difference but does not fully provide the physical explanation in the surrounding text.

Additional CBSE context: Shillong's surrounding relief helps force moisture-bearing winds to rise.

Rising air cools and produces greater rainfall. This is an example of relief or orographic rainfall.

68. WHY DOES DELHI RECEIVE MORE RAINFALL THAN JODHPUR?

The table gives:

- Delhi: **67.0 cm**
- Jodhpur: **36.6 cm** annual rainfall.

The textbook asks students to explain this pattern.

Additional CBSE context: Jodhpur lies in the drier western part of India, while Delhi receives more moisture from monsoon systems. The Aravalli range also runs roughly parallel to the Arabian Sea branch in Rajasthan and therefore does not cause strong uplift there.

69. WHY IS LEH'S CLIMATE SO DIFFERENT?

The table shows that Leh:

- lies at about **3506 m** altitude;
- has winter temperatures below **0°C**;
- receives very little annual precipitation.

This reminds us that climate is controlled not only by latitude but also by factors such as **altitude and relief**.

70. INDIA HAS CLIMATIC DIVERSITY BUT ALSO CLIMATIC UNITY

India has major regional differences:

- some places are extremely wet;
- some are very dry;
- some are very hot;
- some are very cold;
- coastal places have smaller temperature ranges;
- inland places often have larger seasonal temperature ranges.

Yet the textbook asks students to consider whether the **monsoon provides a strong framework of climatic unity** across India.

Main concept

Although different regions experience the monsoon differently, much of India's yearly weather cycle, agriculture, water availability and seasonal life is linked to the monsoon.

Therefore:

Diversity in local climate + Common monsoon influence = Climatic unity

71. HIGH-VALUE CAUSE-EFFECT CHAINS

These are especially useful for weak students because many CBSE questions are based on "Why?", "How?" and "What happens if?"

Temperature and pressure

High temperature → Air becomes warm → Air rises → Low pressure

Low temperature → Air becomes cold → Air sinks → High pressure

Formation of wind

Pressure difference → Air moves from high pressure to low pressure → Wind

Sea breeze

Day → Land heats faster → Low pressure over land → Air moves sea to land → Sea breeze

Land breeze

Night → Land cools faster → Higher pressure over land → Air moves land to sea → Land breeze

South-west monsoon

Summer → India heats → Low pressure over land → High pressure over ocean → Moist winds move sea to land → Rainfall

North-east monsoon

Winter → India cools → High pressure over land → Winds move land to sea → Winds cross Bay of Bengal → Gain moisture → Rainfall on south-eastern coast

Climate change

Human activities → More greenhouse gases → More heat trapped → Temperature rises →

Climate change → Floods/droughts/glacier melt/sea-level rise

72. IMPORTANT DIFFERENCES FOR EXAMS

Weather vs Climate

Basis	Weather	Climate
Meaning	Current atmospheric condition	Long-term average atmospheric condition
Time	Hours or days	Usually decades
Change	Changes quickly	Changes slowly
Example	Rain today	Generally humid climate

Troposphere vs Stratosphere

Troposphere	Stratosphere
Lowest layer	Above troposphere
Average height about 12 km	Extends up to 50 km
Weather occurs here	Few weather disturbances
Most clouds and water vapour	Contains ozone layer
Temperature decreases upward	Suitable for aeroplanes

High Pressure vs Low Pressure

High Pressure	Low Pressure
Associated with colder, sinking air	Associated with warmer, rising air
Often clear weather	Often cloudy/wet weather
Air moves away towards lower pressure	Air receives moving air

Sea Breeze vs Land Breeze

Sea Breeze	Land Breeze
Day	Night
Sea → land	Land → sea
Land warmer	Land cooler
Low pressure over land	Land has cooled more quickly

South-West vs North-East Monsoon

South-West	North-East
Summer monsoon	Winter monsoon
June–September	October–February
Sea → land	Mostly land → sea initially
Moist	Initially dry
Major rainfall for India	Important rainfall for south-east coast

73. IMPORTANT DEFINITIONS

Atmosphere

The blanket of air surrounding Earth.

Gravity

Force of attraction that pulls objects towards Earth.

Altitude

Height of a place above mean sea level.

Weather

Hour-to-hour or day-to-day condition of the atmosphere.

Climate

Average weather conditions over a long period, generally around 30 years or more.

Insolation

Incoming solar energy intercepted by Earth.

Humidity

Presence of water vapour in the air.

Precipitation

Water falling from the atmosphere in forms such as rain, snow, sleet or hail.

Atmospheric Pressure

Pressure caused by the weight of air on Earth's surface.

Wind

Movement of air from high pressure towards low pressure.

Monsoon

Seasonal reversal of wind direction.

Carbon Footprint

Total amount of greenhouse gases released because of human activities.

Climate Change

Long-term changes in weather patterns such as temperature, rainfall and wind.

Biodiversity

Variety of living organisms.

Deforestation

Large-scale removal of forests.

Embankment

A raised structure built along a river to control or prevent flooding.

Silt

Fine soil and mud deposited by flowing water.

74. FACTS THAT SHOULD BE MEMORISED

- Nitrogen — **78%**
- Oxygen — **21%**
- Argon — **0.93%**
- Carbon dioxide — **0.04%**
- Water vapour — generally **0.1–0.4%**
- Troposphere — average height **12 km**

- Stratosphere — up to **50 km**
 - Mesosphere — up to **80 km**
 - Thermosphere — approximately **80–700 km**
 - Weather occurs mainly in — **Troposphere**
 - Ozone layer — **Stratosphere**
 - Meteorites burn — **Mesosphere**
 - Ionosphere and auroras — **Thermosphere**
 - Uppermost layer — **Exosphere**
 - Air pressure is highest — **near sea level**
 - Wind — **high pressure** → **low pressure**
 - Sea breeze — **sea** → **land during day**
 - Land breeze — **land** → **sea during night**
 - India's climate — **Tropical monsoon**
 - South-west monsoon — mainly **June–September**
 - North-east monsoon — mainly **October–February**
 - Monsoon comes from Arabic **mausim**, meaning season.
-

76. CBSE COMPETENCY-BASED UNDERSTANDING

A student should not only memorise facts from this chapter. You should be able to apply them.

Situation 1

The land is very hot but the nearby sea is cooler.

Think: Hot land → rising air → low pressure.

Therefore, air will move from the sea towards land.

Answer: Sea breeze.

Situation 2

A place has cloudy skies and wet weather.

Think: Cloudy and wet weather is often linked with **low pressure**.

Situation 3

An aeroplane wants to avoid strong weather disturbances.

Think: Stratosphere is comparatively free of clouds and major weather disturbances.

Answer: Stratosphere.

Situation 4

A meteor enters Earth's atmosphere.

Think: Meteorites mainly burn in the **mesosphere**.

Situation 5

Farmers receive a weak summer monsoon.

Possible result:

Less rainfall → **water shortage** → **poorer crop growth** → **lower agricultural production** → **effect on livelihoods**

Situation 6

North-east winds cross the Bay of Bengal.

Think: They can gain moisture.

Result:

Rainfall on south-eastern India, especially Tamil Nadu and nearby regions.

Situation 7

A city cuts forests, burns more fossil fuels and increases industrial pollution.

Chain:

More greenhouse gases → more heat trapping → climate change → greater environmental risks.
